

Concept 4 | Applications of Logarithms

English Student Edition • Focused formulas and guided practice

Formula opener

Change base

$$\log_b a = \frac{\ln a}{\ln b}$$

Domain

$$x > 0$$

Inverse

$$b^{\log_b x} = x$$

Classified Practice

Applications of Logarithms

Q001 Written

Solve $\log_2 x = 4$.

Q002 Written**Solve** $\log_2(2x - 1) = \log_2 3$.

Classified Practice

Applications of Logarithms

Q003 **Written**

Solve all 12 given logarithmic equations, including same-base, product, quotient, and shifted-argument forms.

Q004 Written**Solve** $\log_3(x + 2) - \log_9((x - 2)^2) = 1$ **completely.**

Classified Practice

Applications of Logarithms

Q005 Written

Solve $\log_3 x < \log_3 81$ **and** $\log_{1/2}(x + 2) + \log_{1/2}(x - 5) > -3$.

Q006 Written

Solve the six given logarithmic inequalities using base monotonicity and argument conditions.

Classified Practice

Applications of Logarithms

Q007 **Written**

Solve all 24 given logarithmic inequalities, including shifted arguments, coefficients, and rational arguments.

Q008 Written

Solve the **given** challenge inequality involving a reciprocal-base logarithmic sum and state the complete interval.

Classified Practice

Applications of Logarithms

Q009 Written

Solve $5(\log_2 x)^2 - 16 \log_2 x + 3 < 0$.

Q010 Written

Solve the three given substitution problems: two quadratic logarithmic inequalities/equations and one exponential-log value.

Classified Practice

Applications of Logarithms

Q011 Written

Solve the **given** groups: three equations, three inequalities, and three exact logarithmic values.

Q012 Written**Solve** $(\log_3 x)^2 - \log_3 x - 6 < 0$ **for the positive wave-intensity multiplier** x .

Classified Practice

Applications of Logarithms

Q013 Written

For $y = -(\log_3 x)^2 + 4 \log_3 x + 3$, $1 \leq x \leq 27$, find maximum/minimum values and the corresponding x .

Q014 Written

Find the maximum and minimum of the given logarithmic quadratic on its stated interval, with the corresponding values of x .

Classified Practice

Applications of Logarithms

Q015 **Written**

Solve the three given extrema problems, each with a different base and interval.

Q016 Written

For $y = -(\log_2 x)^2 + 2\log_2 x + 4$, $1 \leq x \leq 8$, find the maximum and minimum health ratings and their concentrations.

Classified Practice

Applications of Logarithms

Q017 **Written**

Using $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, determine the digits of 6^{30} and the first non-zero decimal place of $(2/3)^{50}$.

Q018 Written

Complete the given digit-count and first-non-zero-decimal-place sets for the displayed powers and fractions.

Classified Practice

Applications of Logarithms

Q019 Written

Using the printed common logs, determine the number of digits and the leading digit of 12^{50} .

Q020 Written**Solve** $\log_x(x + 2) \leq \log_x(4 - x)$ **for all real** x .

Classified Practice

Applications of Logarithms

Q021 MCQ

The argument condition for $\log_3(x - 5)$ is:**Choose the correct option.**A $x < 5$ B $x \geq 5$ C $x > 5$

D all real x

Q022 MCQ**If $\log_2(2x - 1) = \log_2 3$, then $x =$:****Choose the correct option.****A** 1**B** 2**C** 3**D** 4

Classified Practice

Applications of Logarithms

Q023 MCQ

The valid solution of the challenge equation is:**Choose the correct option.**

A 1

B 2

C 4

D 8

Q024 MCQ**For same-base logarithms, equality of logs gives:****Choose the correct option.**

- A arguments equal
- B bases equal
- C arguments add
- D no result

Classified Practice

Applications of Logarithms

Q025 MCQ

For base $a > 1$, $\log_a M < \log_a N$ means:

Choose the correct option.

A $M > N$

B $M < N$

C $M = N$

D $MN < 0$

Q026 MCQ**For base $0 < a < 1$, the inequality direction is:****Choose the correct option.**

- A preserved
- B reversed
- C removed
- D squared

Classified Practice

Applications of Logarithms

Q027 MCQ

The domain of $\log_{1/2}(x - 5)$ requires:**Choose the correct option.**A $x < 5$ B $x \leq 5$ C $x > 5$ D $x \neq 5$

Q028 MCQ

The warm-up intersection is:**Choose the correct option.**A $0 < x < 5$ B $5 < x < 6$ C $x > 6$ D $-3 < x < 6$

Classified Practice

Applications of Logarithms

Q029 MCQ

The best substitution for a quadratic logarithmic equation is:**Choose the correct option.**

A $t = x^2$

B $t = \log_a x$

C $t = a^x$

D $t = 1/x$

Q030 MCQ

The inequality $5t^2 - 16t + 3 < 0$ gives:**Choose the correct option.**A $t < 1/5$ B $1/5 < t < 3$ C $t > 3$ D $0 < t < 1$

Classified Practice

Applications of Logarithms

Q031 MCQ

The substitution variable must satisfy:**Choose the correct option.**A $t > 0$ B $t \in \mathbb{R}$ C $t < 0$ D $t = 1$

Q032 MCQ

The challenge interval for $(\log_3 x)^2 - \log_3 x - 6 < 0$ is:

Choose the correct option.

A $x < 1/9$

B $1/9 < x < 27$

C $x > 27$

D $0 < x < 3$

Classified Practice

Applications of Logarithms

Q033 MCQ

For $y = -(t - 2)^2 + 7$, the maximum occurs at:

Choose the correct option.

A $t = 0$

B $t = 1$

C $t = 2$

D $t = 3$

Q034 MCQ

The correct first step for extrema is:**Choose the correct option.**

- A differentiate in x
- B set $t = \log_a x$
- C ignore the interval
- D take $x = 0$

Classified Practice

Applications of Logarithms

Q035 MCQ

For $1 \leq x \leq 27$, $t = \log_3 x$ lies in:**Choose the correct option.**A $[-1,3]$ B $[0,3]$ C $[1,27]$

D all real

Q036 MCQ**The water-tank challenge maximum health rating is:****Choose the correct option.****A** 3**B** 4**C** 5**D** 8

Classified Practice

Applications of Logarithms

Q037 MCQ

The number of digits of N is determined by:**Choose the correct option.**

A $\log_{10} N$

B $\lfloor \log_{10} N \rfloor + 1$

C $N+1$

D 10^N

Q038 MCQ**If $-9 < \log_{10} N < -8$, the first non-zero digit is at:****Choose the correct option.**

- A 8th
- B 9th
- C 10th
- D 11th

Classified Practice

Applications of Logarithms

Q039 MCQ

The common logarithm base is:**Choose the correct option.**

A 2

B e

C 10

D any base

Q040 MCQ**The solution set of $\log_x(x + 2) \leq \log_x(4 - x)$ is:****Choose the correct option.****A** (0,1)**B** (1,4)**C** (0,4)**D** empty

Classified Practice

Applications of Logarithms

Quiz MCQ 1 **Concept Quiz · MCQ**

Solve $\log_5(x - 1) = 2$.

Choose the correct option.

A $x = 26$

B 0

C 1

D cannot be determined

Quiz MCQ 2 Concept Quiz · MCQ

For $0 < a < 1$ and $M < N$, compare $\log_a M$ and $\log_a N$.

Choose the correct option.

A $\log_a M > \log_a N$

B 0

C 1

D cannot be determined

Classified Practice

Applications of Logarithms

Quiz MCQ 3 Concept Quiz · MCQ

Solve $4(\log_2 x)^2 - 5\log_2 x + 1 = 0$.**Choose the correct option.**A $x = 2, \sqrt{2}$

B 0

C 1

D cannot be determined

Quiz MCQ 4 Concept Quiz · MCQ

Find the maximum of $y = -(\log_3 x)^2 + 2 \log_3 x + 4$ on $1 \leq x \leq 27$.

Choose the correct option.

A 5 at $x = 3$

B 0

C 1

D cannot be determined

Classified Practice

Applications of Logarithms

Quiz WRITTEN 5 Concept Quiz · Written**Solve** $\log_3(x + 1) = 2$.

Quiz WRITTEN 6 Concept Quiz · Written**Solve** $\log_2 x \geq 3$.

Classified Practice

Applications of Logarithms

Quiz WRITTEN 7 Concept Quiz · Written

Solve $(\log_5 x)^2 - 4 < 0$.

Quiz WRITTEN 8 Concept Quiz · Written

How many digits does 6^{20} have using $\log_{10} 2 = 0.3010, \log_{10} 3 = 0.4771$?

Answer key

Use the question number shown on each page.

- Q001 16
- Q002 $x = 2$
- Q003 Apply the argument condition first; then equate same-base arguments and reject any extraneous roots.
- Q004 $x = 4$
- Q005 $0 < x < 81$; $5 < x < 6$
- Q006 Each solution is the intersection of the argument domain with the monotonicity interval.
- Q007 Use argument conditions, convert coefficients by the power law, then solve and intersect the intervals.
- Q008 Reverse the comparison after converting to a base below one, then intersect with both argument conditions.
- Q009 $2^{1/5} < x < 8$
- Q010 Substitute $t = \log_a x$, solve in t , enforce $x > 0$, and back-substitute.
- Q011 All groups follow the same substitution, domain, solve, and back-substitution sequence.
- Q012 $1/9 < x < 27$
- Q013 Maximum 7 at $x = 9$; minimum 3 at $x = 1$.
- Q014 Use the vertex in $t = \log_a x$ and compare it with both endpoints.
- Q015 For every item, map the domain to t , locate the vertex, and check endpoints.
- Q016 Maximum 5 at $x = 2$; minimum 4 at $x = 1$.
- Q017 24 digits; 9th decimal place.
- Q018 Use $n - 1 < \log_{10} N < n$ for digits and $-n < \log_{10} N < -n + 1$ for the first non-zero place.
- Q019 55 digits; leading digit 1.
- Q020 No real solution.
- Q021 C
- Q022 B
- Q023 C
- Q024 A
- Q025 B
- Q026 B
- Q027 C
- Q028 B
- Q029 B
- Q030 B
- Q031 B
- Q032 B
- Q033 C
- Q034 B
- Q035 B
- Q036 C
- Q037 B
- Q038 B
- Q039 C
- Q040 D
- Quiz MCQ 1 A
- Quiz MCQ 2 B
- Quiz MCQ 3 C
- Quiz MCQ 4 D
- Quiz WRITTEN 5 $x = 8$
- Quiz WRITTEN 6 $x \geq 8$
- Quiz WRITTEN 7 $1/25 < x < 25$
- Quiz WRITTEN 8 16 digits